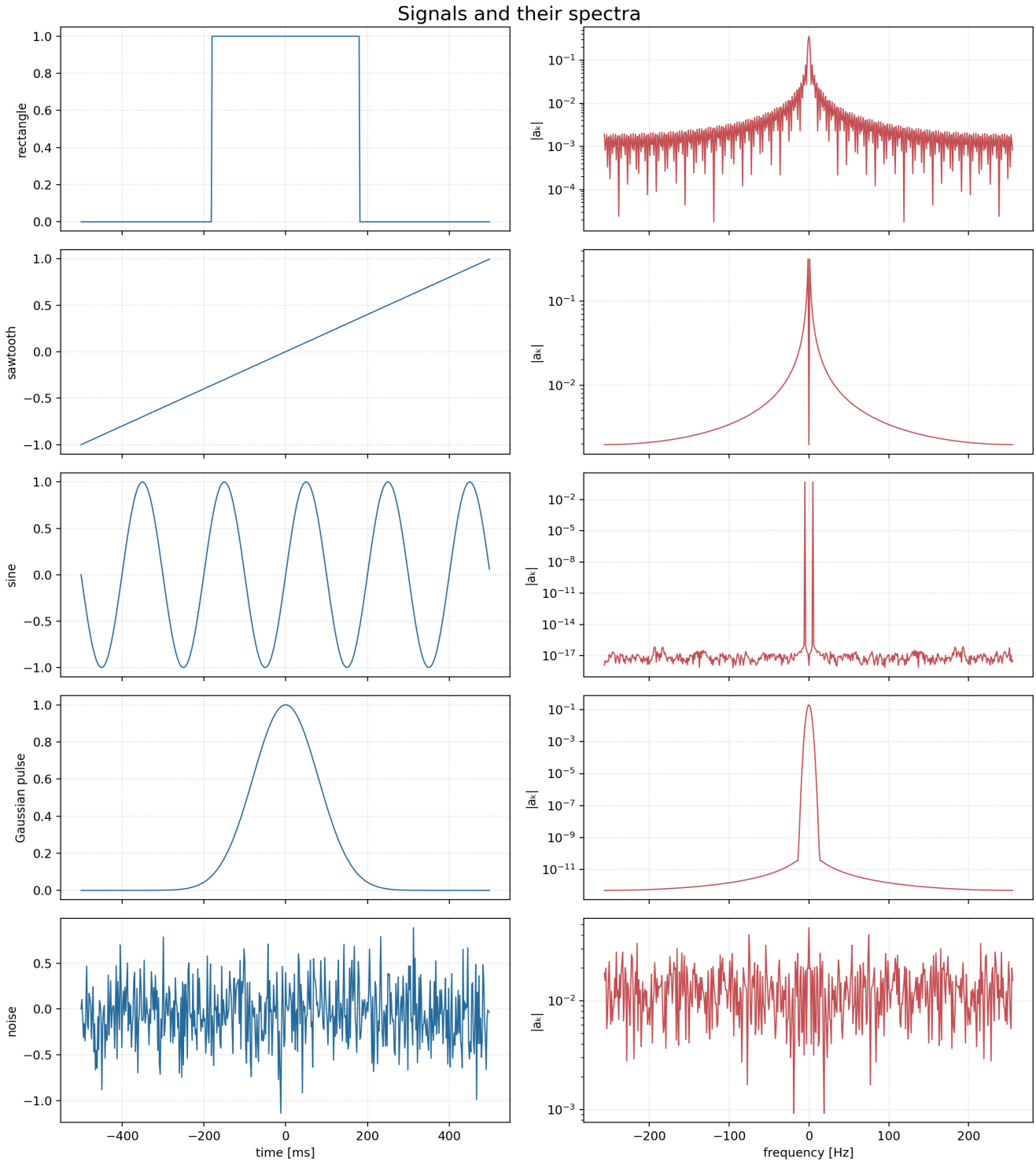
Comparison of Signal Spectra

Parity, smoothness, and finite-window effects

The DFT is computed for a centred, real-valued sample grid. A real signal that is even about the origin has a real spectrum; a real odd signal has an imaginary spectrum; without either parity the spectrum is generally complex. Smooth signals have rapidly decaying coefficients, while jumps create slowly decaying high-frequency tails.

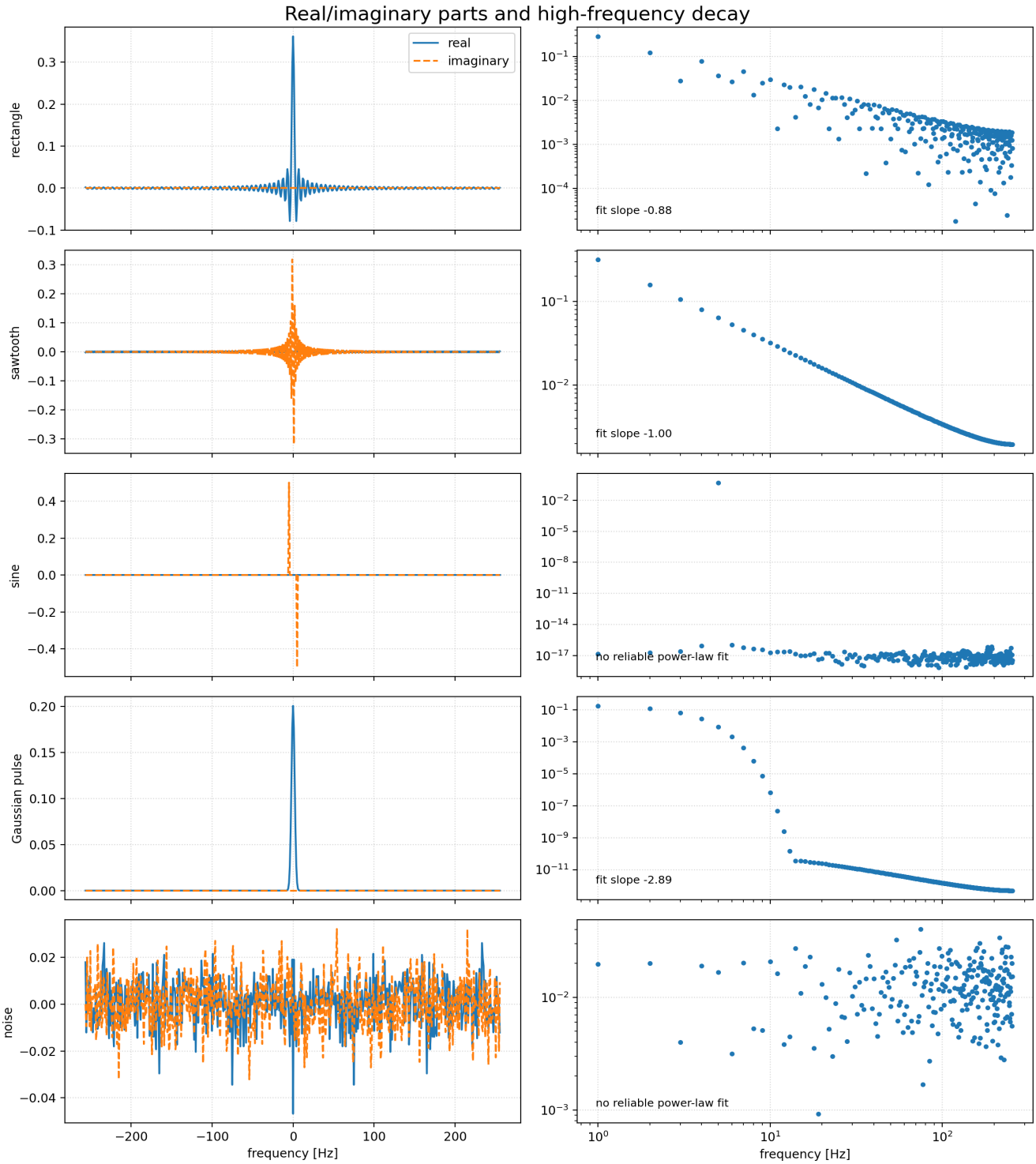
Numerical classification

Signal	Spectrum type	Real / |A| max	Imaginary / |A| m
rectangle	real	1.000	0.000 (slope -0.88)
sawtooth	complex	0.006	1.000 (slope -1.00)
sine	imaginary	0.000	1.000 (slope not fitted)
Gaussian pulse	real	1.000	0.000 (slope -2.89)
noise	complex	1.000	0.687 (slope not fitted)


Signals and their spectra
rectangle
sawtooth
sine
Gaussian pulse
noise
|aₖ|
time [ms]
frequency [Hz]

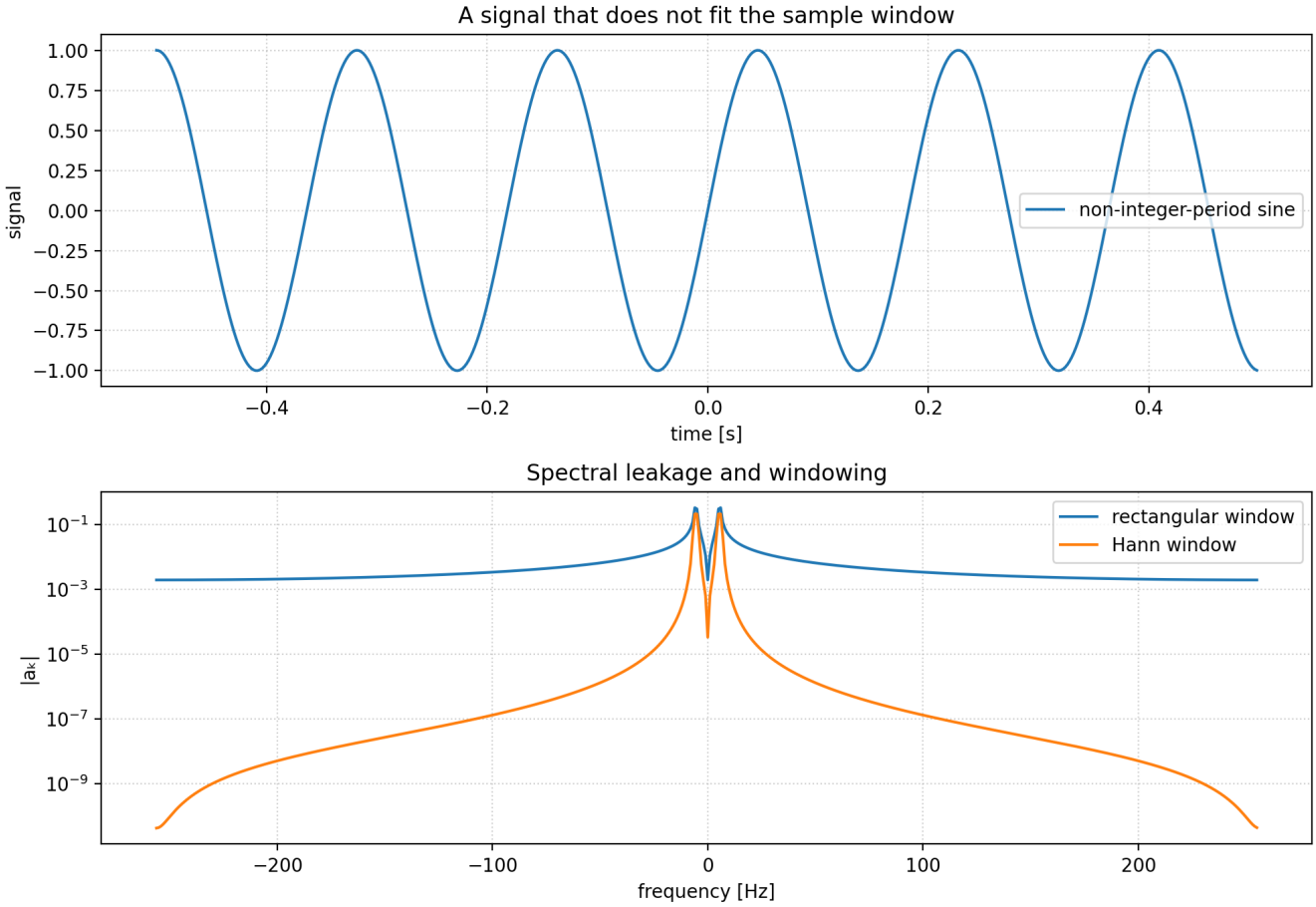
What controls spectral decay?

The rectangle and sawtooth contain discontinuities, so their coefficients have algebraic tails. The sine concentrates energy into a small number of frequencies. The Gaussian is infinitely differentiable and has very rapid, approximately Gaussian decay. White noise has no deterministic smoothness, so its spectrum fluctuates around a broad noise floor and a power-law fit is not physically meaningful.



When the signal does not fit in the sample window

The DFT assumes that the finite sample window repeats periodically. If the first and last samples do not join smoothly, that artificial boundary is a jump. Its Fourier coefficients spread energy into neighbouring bins: this is spectral leakage. A sine with 5.5 cycles in the window is a direct example.



Ways to improve the result: choose a window containing an integer number of periods when possible; increase the observation time to improve frequency resolution; remove a mean or trend; apply a Hann, Hamming, or Blackman window to reduce boundary leakage; and use zero-padding only to interpolate the plotted spectrum. Zero-padding does not add information or remove leakage, while windowing trades reduced sidelobes for a wider main lobe.